BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Programming Challenge and Review

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. A £5-per-ticket program uses total = tickets + 5. What is the correction?

2. Insert 6 into sorted [2, 5, 8]. What is the result?
   A) [2, 5, 8, 6]
   B) [2, 5, 6, 8]
   C) [6, 2, 5, 8]
   D) [8, 6, 5, 2]

3. "A comparison operator must always produce a text paragraph." Correct this. Explain why.

4. A warning needs high temperature OR high pressure. "OR is false when both inputs are true." Correct this. What happens when both are high?

Today's objectives

Independently design and develop a small program. Apply the programming techniques to a new problem.

Key vocabulary - Read, cover, recall

Planning
Define inputs, processing, output and success criteria.

Selection
Use IF or ELSE to choose valid outcomes.

Iteration
Repeat actions with a bounded FOR or condition-controlled WHILE.

Testing
Check normal, boundary and invalid inputs, then refine.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Planning
Define inputs, processing, output and success criteria. For example, consider a club booking tool.

Selection
Use IF or ELSE to choose valid outcomes. For example, consider a revision calculator.

Iteration
Repeat actions with a bounded FOR or condition-controlled WHILE. For example, consider a rewards app.

Testing
Check normal, boundary and invalid inputs, then refine. For example, consider a canteen voucher.

I Do - Worked model

A club booking tool is being developed. Explain how planning and selection are used. [4 marks]

- Identify where this particular program needs control or values.
- Planning [1]; Define inputs, processing, output and success criteria. [1].
- Selection [1]; Use IF or ELSE to choose valid outcomes. [1].
- Give one accurate point and a linked reason for each technique.

We Do - Guided practice

1. A revision calculator is being programmed. Explain why planning matters. [2 marks]

2. For a rewards app, explain how selection is used. [2 marks]

3. A canteen voucher needs iteration and testing. Explain the roles of both. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Write the missing condition to report a hot sensor.

```
sensor = 29
if ... then
print("Too hot")
else
print("OK")
endif
```

You Do - Independent exam questions

Q2. State TWO concepts needed for this programming technique. [2 marks]

Q3. Explain how selection and iteration can help a fundraising tickets. [4 marks]

Q4. Explain the roles of planning, iteration and testing when creating a quiz score. [6 marks]

Exit Ticket - True or False?

A useful algorithm plan identifies inputs, processing, outputs and success criteria.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Use total = tickets * 5; addition does not calculate £5 for each ticket.

2. B - [2, 5, 6, 8]

3. A Boolean result: true or false.

4. OR is true if at least one input is true, including both. AND needs both true. NOT reverses a value.

I Do - Explain the model

Identify where this particular program needs control or values.

Planning [1]; Define inputs, processing, output and success criteria. [1].

Selection [1]; Use IF or ELSE to choose valid outcomes. [1].

Give one accurate point and a linked reason for each technique.

We Do - Model answers

1. Planning [1]; Define inputs, processing, output and success criteria. [1].

2. Selection [1]; Use IF or ELSE to choose valid outcomes. [1].

3. Iteration [1]; Repeat actions with a bounded FOR or condition-controlled WHILE. [1]. Testing [1]; Check normal, boundary and invalid inputs, then refine. [1].

Code First - Answer

```
sensor = 29
if sensor > 28 then
print("Too hot")
else
print("OK")
endif
```

Expected output: Too hot

You Do - Self-mark

Q2: Planning [1]; Iteration [1].

Q3: Selection [1]; Use IF or ELSE to choose valid outcomes. [1]. Iteration [1]; Repeat actions with a bounded FOR or condition-controlled WHILE. [1].

Q4: Planning [1]; Define inputs, processing, output and success criteria. [1]. Iteration [1]; Repeat actions with a bounded FOR or condition-controlled WHILE. [1]. Testing [1]; Check normal, boundary and invalid inputs, then refine. [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.